

Problem 2: (10 points) Bayes Classifiers

Consider the table of measured data given at right. We will use the three observed features x_1, x_2, x_3 to predict the class y . In the case of a tie, we will prefer to predict class $y = 0$.

x_1	x_2	x_3	y
0	0	0	1
1	1	0	1
1	0	1	1
1	0	1	1
0	1	0	0
0	0	1	0
1	1	1	0
0	1	0	0

(a) Write down the probabilities used by a naïve Bayes classifier:

$$p(y = 0) : \frac{1}{2}$$

$$p(y = 1) : \frac{1}{2}$$

$$p(x_1 = 1|y = 0) : \frac{1}{4}$$

$$p(x_1 = 1|y = 1) : \frac{3}{4}$$

$$p(x_1 = 0|y = 0) : \frac{3}{4}$$

$$p(x_1 = 0|y = 1) : \frac{1}{4}$$

$$p(x_2 = 1|y = 0) : \frac{3}{4}$$

$$p(x_2 = 1|y = 1) : \frac{1}{4}$$

$$p(x_2 = 0|y = 0) : \frac{1}{4}$$

$$p(x_2 = 0|y = 1) : \frac{3}{4}$$

$$p(x_3 = 1|y = 0) : \frac{1}{2}$$

$$p(x_3 = 1|y = 1) : \frac{1}{2}$$

$$p(x_3 = 0|y = 0) : \frac{1}{2}$$

$$p(x_3 = 0|y = 1) : \frac{1}{2}$$

(b) Using your naïve Bayes model, compute:

$$p(y = 0|x_1 = 1, x_2 = 0, x_3 = 0) :$$

$$p(y = 1|x_1 = 1, x_2 = 0, x_3 = 0) :$$

$$= \frac{(\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{2})}{\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}}$$

$$= 1 - .9 = .1$$

$$= \frac{9}{9+1} = .9$$

(c) Give an example of a problem setting in which we might want to use naïve Bayes even though the model assumptions may not be correct.

Problems with many features, such as text using word appearance or counts
(eg, spam filtering, as a classic example).

Problem 3: (10 points) Decision Trees

Consider the table of measured data given at right. (Note that some data points are repeated.) We will use a decision tree to predict the outcome y using the three features, $x_1, \dots, x_3$. In the case of ties, we prefer to use the feature with the smaller index (x_1 over x_2 , etc.) and prefer to predict class 1 over class 0. You may find the following values useful (although you may also leave logs unexpanded):

$$\log_2(1) = 0 \quad \log_2(2) = 1 \quad \log_2(3) = 1.59 \quad \log_2(4) = 2 \\ \log_2(5) = 2.32 \quad \log_2(6) = 2.59 \quad \log_2(7) = 2.81 \quad \log_2(8) = 3$$

x_1	x_2	x_3	y
1	1	1	1
1	0	0	1
1	0	0	1
0	0	0	0
0	1	1	0
1	0	1	0

(a) What is the entropy of y ?

$$p(y) = 1/2 \Rightarrow H(y) = 1/2 \log 2 + 1/2 \log 2 = 1 \text{ bit.}$$

(b) Which variable would you split first? Justify your answer.

x_1 has highest info gain

$$x_1: IG = H(1/2) - 2/3 \cdot H(3/4) - 1/3 \cdot H(0)$$

$$x_2: IG = H(1/2) - 2/3 H(1/2) - 1/3 H(1/2) \quad \times$$

$$x_3: IG = H(1/2) - 1/2 H(2/3) - 1/2 H(1/3)$$

$$(H(0) < H(3/4) < H(2/3) < H(1/2))$$

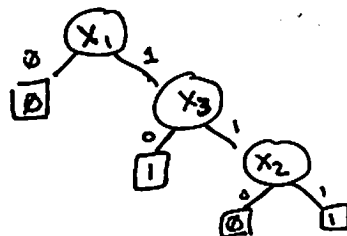
(c) What is the information gain of the variable you selected in part (b)?

$$IG = 1 - 2/3 H(3/4) - 1/3 H(0)$$

$$H(3/4) = 3/4 \log 4/3 + 1/4 \log 4$$

$$H(0) = 0$$

(d) Draw the rest of the decision tree learned on these data.

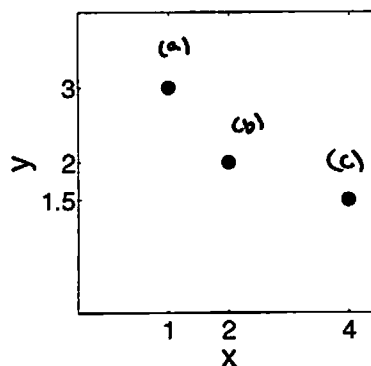


$\leftarrow x_3$ is better than x_2

(intuitively, $3/4$ it gets more points correct, but you can evaluate IG also)

Problem 4: (12 points) Nearest Neighbor Regression

Consider the data points shown at right, for a regression problem to predict y given a scalar feature x . We wish to use a k -nearest neighbor learner to minimize the mean squared error (MSE) of our predictions.



(a) Compute the training error (MSE) of a 1-nearest neighbor predictor.

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(b) Compute the leave-one-out cross-validation error (MSE) of a 1-nearest neighbor predictor.

$$\begin{array}{ll} \text{leave out (a)} \Rightarrow \text{predict } 2 \Rightarrow (1)^2 \\ \text{(b)} \quad \quad \quad 3 \Rightarrow (1)^2 \\ \text{(c)} \quad \quad \quad 2 \Rightarrow (\frac{1}{2})^2 \end{array} \quad \text{MSE} = \frac{1}{3} \left[1 + 1 + \frac{1}{4} \right] = \frac{1}{3} \cdot \frac{9}{4} = \frac{3}{4}.$$

(c) Compute the training error (MSE) of a 2-nearest neighbor predictor.

$$\begin{array}{ll} \text{predict for (a)} \Rightarrow 2.5 \\ \text{(b)} \Rightarrow 2.5 \\ \text{(c)} \Rightarrow 1.75 \end{array} \quad \Rightarrow \frac{1}{3} \left[\frac{1}{4} + \frac{1}{4} + \frac{1}{16} \right] = \frac{3}{16}.$$

(d) Compute the leave-one-out cross-validation error (MSE) of a 2-nearest neighbor predictor.

$$\begin{array}{ll} \text{LO (a)} \Rightarrow \text{predict } 1.75 \\ \text{(b)} \Rightarrow 2.25 \\ \text{(c)} \Rightarrow 2.5 \end{array} \quad \Rightarrow \frac{1}{3} \left[\left(\frac{5}{4}\right)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{3}{4}\right)^2 \right] = \frac{35}{3 \cdot 16} = \frac{35}{48}$$

(e) Based on just these data & results, which value of k would you choose, and why?

$k=2$ better predicted
LOO CV than $k=1$
(barely...)

$$\begin{array}{ll} k=1 \Rightarrow \text{LOO CV} = \frac{36}{48} \\ k=2 \Rightarrow \text{LOO CV} = \frac{35}{48} \end{array}$$

Problem 5: (10 points) Multiple Choice

For the following questions, assume that we have m data points $y^{(i)}, x^{(i)}, i = 1 \dots m$, each with n features, $x^{(i)} = [x_1^{(i)} \dots x_n^{(i)}]$.

True or false: The predictions of a decision tree classifier will not be affected if we pre-process the data to normalize the magnitude of each feature (e.g., rescale each feature to the range $[-1,1]$).

True or false: Linear regression can be solved using either matrix algebra or gradient descent.

True or false: Using backpropagation to train a neural network will avoid getting stuck in local optima.

True or false: With sufficient depth, a decision tree can approximate any boolean function.

True or false: If two models have the same VC dimension, they are equally likely to overfit the data.

True or false: Given sufficiently many data m , the 1-nearest neighbor classifier error rate approaches the Bayes optimal error rate.

True or false: Stochastic gradient descent is often preferred over batch when the number of data points m is very large.

True or false: Increasing the regularization penalty of a linear regression model will decrease its bias.

A quadratic program (such as SVM learning) is characterized by a quadratic objective function and a set of convex linear quadratic soft constraints.

Suppose that we are training a linear regressor (perceptron). Adding extra features that are not related to the target (e.g., randomly generated values) before training will typically increase not change decrease our training error.